CHEST

Refining the Lung Allocation Score Models Fails to Improve Discrimination Performance

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BACKGROUND: As broader geographic sharing is implemented in lung transplant allocation through the Composite Allocation Score (CAS) system, models predicting waitlist and posttransplant (PT) survival will become more important in determining access to organs.

RESEARCH QUESTION: How well do CAS survival models perform, and can discrimination performance be improved with alternative statistical models or machine learning approaches?

STUDY DESIGN AND METHODS: Scientific Registry for Transplant Recipients (SRTR) data from 2015-2020 were used to build seven waitlist (WL) and data from 2010-2020 to build similar PT models. These included the (I) current lung allocation score (LAS)/CAS model; (II) re-estimated WL-LAS/CAS model; (III) model II incorporating nonlinear relationships; (IV) random survival forests model; (V) logistic model; (VI) linear discriminant analysis; and (VII) gradient-boosted tree model. Discrimination performance was evaluated at 1, 3, and 6 months on the waiting list and 1, 3, and 5 years PT. Area under the curve (AUC) values were estimated across subgroups.

RESULTS: WL model performance was similar across models with the greatest discrimination in the baseline cohort (AUC 0.93) and declined to 0.87-0.89 for 3-month and 0.84-0.85 for 6-month predictions and further diminished for residual cohorts. Discrimination performance for PT models ranged from AUC 0.58-0.61 and remained stable with increasing forecasting times but was slightly worse for residual cohorts. WL and PT variability in AUC was greatest for individuals with Medicaid insurance.

INTERPRETATION: Use of alternative modeling strategies and contemporary cohorts did not improve performance of models determining access to lung transplant.

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ABBREVIATIONS: ADI = Area deprivation index; AUC = area under receiver operating characteristic curve; CAS = composite allocation score; LAS = lung allocation score; PT = posttransplant; STRT = Scientific Registry for Transplant Recipients; WL = waitlist

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Take-home Points

Study Question: How well do CAS survival models perform, and can discrimination performance be improved with alternative statistical models or machine learning approaches?

Results: Waitlist model AUC was excellent and similar among models with the highest discrimination performance in the baseline cohort; PT model AUC was poor, but discrimination performance did not degrade with increasing time from transplant.

Interpretation: Use of advanced regression and machine learning approaches on contemporary cohorts of lung transplant candidates and recipients did not improve performance of models determining access to lung transplant in the United States.

Lung transplant candidates in the United States are prioritized for transplant based on their estimated transplant benefit represented by the Lung Allocation Score (LAS).[1] The LAS is computed using two Cox proportional hazards models—one for likelihood of waitlist (WL) survival in the next year and another for likelihood of posttransplant (PT) survival in the next year. In current allocation policy, the WL model is given twice the weight of the PT survival model. The LAS has undergone few structural changes since its implementation in 2005, including the addition of disease-specific variables for pulmonary vascular disease in 2015, and a recent cohort update and coefficient reparameterization implemented in September 2021.

The US lung transplant allocation system is poised to move to the composite allocation score (CAS) system,[2,3] in which geography plays a less important role and there is broader geographic sharing of donor organs. In this system, estimates of WL and PT survival will play greater roles in determining patient access to transplant. The CAS incorporates WL survival over 1 year and PT survival over 5 years, providing each with equal weight in its overall calculation; however, the WL mortality model assigns higher relative weight to WL mortality for those at very high risk of WL mortality in the overall calculation.

Therefore, the goals of this study were to (1) study *discrimination* performance of the CAS survival models—in other words, the ability of the models to effectively separate patients with higher likelihood of mortality from patients with lower likelihood of mortality; (2) compare CAS survival model performance with that of alternative statistical modeling and machine learning approaches, which relax assumptions of linearity of relationships and further allow for interaction effects among risk factors; and (3) evaluate the extent to which model performance varies by socioeconomic and clinical factors.

Study Design and Methods

Data Sources and Inclusion Criteria

We analyzed retrospective data on US lung transplant candidates aged ≥ 18 years. This study used data from the Scientific Registry of Transplant Recipients (SRTR). The SRTR data system includes data on all donor, waitlisted candidates, and transplant recipients in the United States, submitted by the members of the Organ Procurement and Transplantation Network (OPTN). The Health Resources and Services Administration (HRSA), US Department of Health and Human Services provides oversight to the activities of the OPTN and SRTR contractors. The WL survival cohort began February 19, 2015, and the PT cohort began January 1, 2010; both ended May 18, 2020. Those with missing data on age, sex, supplemental oxygen, or mechanical respiratory support were excluded. Those who were listed more than once were reconciled to only have one observation with their earliest listing date. If retransplanted in the cohort, individuals were censored for PT analyses at the time of their second transplant. The waitlist and PT survival analyses included variables and coefficients that will be used for the CAS survival models, including the 1-year waitlist model already implemented in November 2021, and the future 5-year PT model that will go into effect with implementation of the CAS system (e-Appendix 1).[4] Primary outcomes included (1) time from WL placement to death or removal from the waiting list because of the status "too sick for transplant" (codes 8 and 13) and (2) time from transplantation to death or retransplantation. The "too sick for transplant" codes are entered by individual transplant centers and are based on clinical judgment. This study was approved by the Cleveland Clinic Institutional Review Board (20-373). The data reported here have been supplied by the Hennepin Healthcare Research Institute (HHRI) as the contractor for the SRTR. The interpretation and reporting of these data are the responsibility of the authors and in no way should be seen as an official policy of or interpretation by the SRTR or the US Government.

Model Development

We evaluated four models for each of the two primary outcomes. For waitlist mortality, we considered the model currently in use for LAS and to be used for CAS model (WL-LAS/CAS, implemented November 2021), which is a weighted Cox proportional hazards regression model (model I); a re-estimated WL-LAS/CAS model incorporating the most up-to-date data through May 26, 2020, resulting in updated cohort and parameterizations (model II); a modification of model II that incorporated nonlinear relationships for continuous predictors via cubic splines (model III); and a

random survival forests model (model IV). To test other machine learning techniques, we used survival stacking to convert the primary outcome into a binary response.[5] Survival stacking is an approach used to convert censored time-to-event data into a format amenable to applying techniques for binary outcomes.[5] We used a logistic model (model V), linear discriminant analysis (model VI), and a gradient boosted tree (model VII). The WL mortality model was stacked in 2-week increments. We analyzed PT survival in a similar fashion, with model I representing the CAS PT survival model (PT-CAS), and the periods were stacked using 6-month increments.

Data were randomly partitioned into 10 training and test samples for cross-validation. WL models were estimated among (1) all candidates who met the above-listed inclusion criteria; (2) candidates who remained on the waiting list at 3 months, and (3) candidates who remained on the waiting list at 6 months. PT models were estimated among: (1) all transplant recipients using time of transplant as baseline and (2) residual cohorts of transplant recipients who survived 1, 3, and 5 years after transplant. Data in the residual cohorts used a candidate's most recent characteristics nearest to the time point measured. We used the inverse probability-of-censoring weighting method to account for informative censoring attributable to transplant (via the R package *pec*[6]) and multiple imputation to account for missing predictor data. Further details on modeling, including software and reproducibility methods, are provided in e-Appendix 1.

Model Validation

We evaluated predictive accuracy of all developed models using procedures aligned with the Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD) guidelines for internal model validation. Discrimination performance was evaluated by comparing forecasted waitlist survival rates at 1, 3, and 6 months from each cohort's baseline time and PT survival rates at transplant, 1, 3, and 5 years PT using area under receiver operating characteristic curve (AUC; An AUC of 1.0 indicates a risk model can perfectly separate those who have survived from those who have died, and an AUC of 0.5 indicates a risk model produces the same distribution of predicted risk for those who have died and those who have survived [ie, no better than chance]) estimates, using the *riskRegression* R package.[1,7] AUC values were estimated within levels of the following demographic and socioeconomic factors: age, sex, race/ethnicity, area deprivation index (ADI), insurance coverage, education, and diagnosis group (The four diagnosis groups include Group A, obstructive lung disease; Group B, pulmonary vascular disease; Group C, cystic fibrosis and immunodeficiency disorders; and Group D, restrictive lung diseases)[2] for transplant candidates (at time of waitlisting) and recipients (at time of transplant). ADI was estimated based on candidates' zip code tabulation area at time of waitlisting, using the *sociome* R package.[8] As a sensitivity analysis, to compare model I, model II, and model III, we calculated Uno's C statistic using the *survAUC* R package.[9] This reflects the concordance index over the whole interval for a forecasted time point.

Results

Waitlist Survival

Of 13,753 candidates placed on the waiting list during the study period, 18 were removed for missing data on age, respiratory support, or supplemental oxygen values, resulting in a total of 13,735 candidates. After reconciling those who were listed more than once, 13,204 candidates were included for analysis. Four thousand nine hundred twenty-nine candidates remained on the waiting list after 3 months, 2,078 after 6 months, and 1,202 after 1 year (Table 1). Candidates in group A represented 24% of the WL population at listing and increased to 43% in the residual cohort remaining on the waiting list at 1 year. Candidates in group D represented 61% of the WL population at time of listing and declined to 42% in the residual cohort at 1 year. The relative percentages for candidates in groups B and C remained similar from baseline to 1 year, 5.5% to 6.2% and 8.9% to 9.5%, respectively. The residual cohort of candidates who remained on the waiting list for at least 1 year were more likely to be female, less likely to have a 6-min walk distance of zero (observed in the sickest candidates), and less likely to require oxygen at rest compared with the initial population at listing.

AUC measurements for all models estimating WL mortality are provided in Table 2. All WL models performed similarly to one another within each cohort and over varying time horizons. In the baseline cohort, model I (WL-LAS/CAS) had a median AUC of 0.92 for survival predictions at 1 month, whereas the other models each had a median AUC of 0.93, except for model VII, which had a median AUC value of 0.92. Across models, AUC values declined as the baseline cohort spent more time on the WL (ie, AUC values ranging from 0.87 to 0.89 for 3-month survival predictions from time of waitlisting and from 0.84 to 0.85 for 6-month survival predictions).

Discrimination performance diminished when models estimated mortality for residual cohorts of WL candidates remaining on the list at 3 months, 6 months, and 12 months compared with estimates for candidates at listing. For the cohort who remained on the waiting list at 3 months, at a forecasted time of 1 month from the start of the cohort (4 months from placement on the waiting list), model I had an AUC of 0.85 compared with 0.84 for model II, 0.79 for model III, 0.83 for model IV, 0.85 for model V, 0.82 for model VI, and 0.77 for model VII. At a 3-month forecasted time (6 months from placement on the waiting list), the AUCs ranged from 0.72 to 0.79. At a 6-month forecasted time, AUCs ranged from 0.73 to 0.78. For the 6-month cohort, discrimination performance was lower than 1- and 3-month cohorts at a 1-month forecasted time but was similar to the 3-month cohort's discrimination at 3 and

TABLE 1] Candidate Characteristics by Time on the Waiting List

Characteristic	Baseline	3 Mo	6 Mo	12 Mo
No. at risk	N = 13,204[a]	n = 4,929[a]	n = 2,708[a]	n = 1,202[a]
No. of deaths		n = 781	n = 209	n = 192
No. censored		n = 7494	n = 2012	n = 1314
Diagnosis group				
A	3,229 (24%)	1,684 (34%)	1,046 (39%)	514 (43%)
B	729 (5.5%)	281 (5.7%)	164 (6.1%)	75 (6.2%)
C	1,171 (8.9%)	490 (9.9%)	271 (10%)	114 (9.5%)
D	8,075 (61%)	2,474 (50%)	1,227 (45%)	499 (42%)
6-min walk distance, feet				
0	714 (5.5%)	65 (1.3%)	25 (0.9%)	7 (0.6%)
1-500	2,708 (21%)	747 (15%)	391 (14%)	174 (14%)
501-1,000	5,405 (41%)	2,222 (45%)	1,238 (46%)	570 (47%)
1,001-1,500	3,825 (29%)	1,694 (34%)	950 (35%)	406 (34%)
1,500+	444 (3.4%)	198 (4.0%)	104 (3.8%)	45 (3.7%)
Not assessed	108	3		
Respiratory support				
Mechanical ventilation	718 (5.4%)	99 (2.0%)	46 (1.7%)	16 (1.3%)
BiPAP	960 (7.3%)	403 (8.2%)	249 (9.2%)	136 (11%)
CPAP	802 (6.1%)	323 (6.6%)	202 (7.5%)	78 (6.5%)
None	10,724 (81%)	4,104 (83%)	2,211 (82%)	972 (81%)
Systolic pulmonary Arterial pressure	40 (32, 50)	37 (31, 46)	37 (31, 46)	36 (31, 46)
Not assessed	2,612	246	47	19
Supplemental oxygen, L/min	4.0 (3.0, 6.0)	4.0 (2.0, 6.0)	4.0 (2.5, 6.0)	4.0 (2.0, 6.0)
Oxygen use				
At rest	11,339 (86%)	3,983 (81%)	2,170 (80%)	951 (79%)
At night	204 (1.5%)	107 (2.2%)	53 (2.0%)	23 (1.9%)
With exercise	1,320 (10.0%)	674 (14%)	395 (15%)	172 (14%)
None	341 (2.6%)	165 (3.3%)	90 (3.3%)	56 (4.7%)
Performs ADLs without assistance	943 (7.1%)	459 (9.3%)	266 (9.8%)	112 (9.3%)
Creatinine	0.80 (0.69, 0.97)	0.80 (0.67, 0.94)	0.80 (0.67, 0.94)	0.79 (0.64, 0.91)
Not assessed	22			
BMI	26.1 (22.3, 29.3)	25.7 (21.7, 29.1)	25.6 (21.7, 29.0)	25.5 (21.9, 28.9)
Not assessed	3			
Bilirubin	0.50 (0.30, 0.70)	0.40 (0.30, 0.60)	0.40 (0.30, 0.60)	0.40 (0.30, 0.50)
Not assessed	151	17	3	
Female sex	5,442 (41%)	2,522 (51%)	1,538 (57%)	738 (61%)
Race/ethnicity				
Hispanic	1,209 (9.2%)	434 (8.8%)	256 (9.5%)	115 (9.6%)
Non-Hispanic Black	1,327 (10%)	540 (11%)	314 (12%)	141 (12%)
Non-Hispanic White	10,215 (77%)	3,779 (77%)	2,037 (75%)	894 (74%)
Other	453 (3.4%)	176 (3.6%)	101 (3.7%)	52 (4.3%)
Area deprivation index Quintile				
Q1 (Lowest deprivation)	4,255 (32%)	1,631 (33%)	889 (33%)	371 (31%)
Q2	2,548 (19%)	957 (20%)	534 (20%)	238 (20%)

(Continued)

TABLE 1] *(Continued)*

Characteristic	Baseline	3 Mo	6 Mo	12 Mo
Q3	2,390 (18%)	858 (18%)	448 (17%)	207 (17%)
Q4	2,081 (16%)	752 (15%)	437 (16%)	203 (17%)
Q5 (Highest deprivation)	1,839 (14%)	695 (14%)	386 (14%)	175 (15%)
Not assessed	91	36	14	8
Insurance				
Private	6,082 (46%)	2,295 (47%)	1,241 (46%)	523 (44%)
Medicare	5,514 (42%)	2,036 (41%)	1,130 (42%)	516 (43%)
Medicaid	1,114 (8.4%)	424 (8.6%)	242 (8.9%)	119 (9.9%)
Other	494 (3.7%)	174 (3.5%)	95 (3.5%)	44 (3.7%)
Education				
High school or less	5,228 (41%)	2,049 (43%)	1,168 (44%)	551 (47%)
College/technical school	6,255 (49%)	2,317 (48%)	1,223 (47%)	516 (44%)
Postgraduate	1,307 (10%)	428 (8.9%)	239 (9.1%)	100 (8.6%)
Not assessed	414	135	78	35

Time points assessed include at the time of placement on the waiting list, 1 months, 3 months, and 6 months. ADLs = activities of daily living; group A = obstructive lung disease; group B = pulmonary vascular disease; group C = cystic fibrosis and immunodeficiency disorders; group D = restrictive lung diseases.
[a]No. (%); Median (interquartile ratio).

6 months of forecasted time. Reestimated hazard ratios for the WL mortality model are presented in e-Table 1.

PT Survival

Twenty thousand seven hundred sixty-three adult transplant recipients were included for analysis of PT survival. After 1 year, 2,494 transplant recipients died and 3,123 were censored because they were lost to follow-up, leaving 15,146 in the 1-year cohort, 8,588 in the 3-year cohort, and 4,358 in the 5-year cohort (Table 3).

AUC measurements for all models estimating PT survival are provided in Table 4. Overall, the discrimination performance of model I (PT-CAS) was only slightly better than guessing at most time points. Compared with model I, discrimination performance was similar across models (0.58-0.61) and remained relatively stable with increasing forecasting times (0.58-0.60) in the baseline cohort.

In the residual cohort of recipients who survived for at least 1 year, discrimination performance was slightly worse for all models at 1 year of forecasted time (median AUC values of 0.59 [model I], 0.57 [model II], 0.58 [model III)], 0.57 [model IV], 0.56 [model V], 0.56 [model VI], 0.57 [model VII]) as compared with the performance of models originating at time of transplant using the entire cohort of transplant recipients. This pattern was similar within the 3- and 5-year residual cohorts (Table 4). Models I through III were also compared with Uno's C statistic for all WL and PT residual cohorts and forecasted survival time points. All models performed comparatively regardless of residual cohort and survival time (e-Tables 2, 3).

Subgroup Analyses

Figure 1A shows the AUC values for 6-month WL survival estimates from the time of waitlisting within subgroups defined according to demographic and socioeconomic factors. Models had the worst performance for diagnosis group A with the lowest median AUC at 0.74, as compared with values of 0.83 or higher for groups B, C, and D. Discrimination performance of models was higher for males compared with females (median AUC, 0.87 and 0.83, respectively). Finally, AUC model performance was lowest with greater variability across the 10 validation folds for those that were on Medicaid, when compared with Medicare and those who were privately insured.

Figure 1B presents stratified model discrimination analyses for PT survival with a 5-year time horizon. e-Figure 1 displays this on a 1-year time horizon. For each of the subgroups, discrimination performance of the models was generally worse for PT survival as

TABLE 2] Discrimination Performance of Waitlist Mortality Models

	Baseline Cohort						
Forecasted Time	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII
1 Mo	0.92 (0.88-0.97)	0.93 (0.88-0.96)	0.93 (0.88-0.96)	0.93 (0.88-0.96)	0.93 (0.89-0.96)	0.93 (0.87-0.97)	0.92 (0.87-0.96)
3 Mo	0.88 (0.85-0.89)	0.89 (0.85-0.9)	0.89 (0.84-0.91)	0.88 (0.87-0.9)	0.89 (0.84-0.90)	0.88 (0.84-0.91)	0.87 (0.84-0.9)
6 M	0.84 (0.83-0.88)	0.85 (0.83-0.89)	0.85 (0.83-0.9)	0.85 (0.81-0.89)	0.85 (0.83-0.90)	0.85 (0.80-0.90)	0.84 (0.81-0.89)
	3-Mo Cohort						
Forecasted Time	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII
1 Mo	0.85 (0.58-0.92)	0.84 (0.56-0.92)	0.79 (0.54-0.9)	0.83 (0.64-0.96)	0.85 (0.52-0.94)	0.82 (0.6-0.94)	0.77 (0.59-0.89)
3 Mo	0.77 (0.64-0.93)	0.77 (0.65-0.92)	0.77 (0.68-0.89)	0.77 (0.64-0.93)	0.79 (0.67-0.95)	0.79 (0.64-0.95)	0.72 (0.63-0.91)
6 Mo	0.76 (0.72-0.81)	0.77 (0.73-0.83)	0.76 (0.72-0.83)	0.77 (0.69-0.84)	0.78 (0.74-0.82)	0.78 (0.71-0.84)	0.73 (0.62-0.87)
	6-Mo Cohort						
Forecasted Time	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII
1 Mo	0.79 (0.58-0.96)	0.79 (0.71-0.96)	0.81 (0.44-0.94)	0.72 (0.33-0.97)	0.79 (0.71-0.88)	0.77 (0.74-0.93)	0.74 (0.59-0.91)
3 Mo	0.77 (0.68-0.96)	0.82 (0.62-0.91)	0.78 (0.68-0.94)	0.81 (0.62-0.88)	0.81 (0.63-0.92)	0.82 (0.61-0.93)	0.78 (0.63-0.95)
6 Mo	0.73 (0.69-0.85)	0.77 (0.64-0.87)	0.76 (0.62-0.83)	0.75 (0.64-0.86)	0.79 (0.66-0.85)	0.79 (0.66-0.86)	0.71 (0.56-0.81)

Median AUC (Min, Max). Model I: WL-CAS; Model II: Re-estimation of WL-CAS to the specified residual cohort with an updated cohort and parameterizations; Model III: Cox regression model with cubic splines accounting for nonlinear relationships; Model IV: Random survival forest model; Model V: A logistic model calculated using the stacked survival method; Model VI: Linear discriminant analysis; Model VII: Gradient boosted tree model. The forecasted time ahead indicates the time from the cohort-defined period. For example, forecasted time ahead of 1 month from the baseline cohort would be 1 month in actual time, whereas 1 month from the 3-month cohort would be 4 months of actual time. CAS = composite allocation score, WL = waitlist.

TABLE 3] Transplant Recipient Characteristics by Time After Transplant

Characteristic	Time of Transplant	1 Y	3 Y	5 Y
No. at risk	N = 20,763[a]	n = 15,146[a]	n = 8,588[a]	n = 4,358[a]
No. of deaths		n = 2,494	n = 2,914	n = 1,720
No. censored		n = 3,123	n = 3,644	n = 2,330
Diagnosis group				
A	5,633 (27%)	4,287 (28%)	2,522 (29%)	1,343 (30%)
B	822 (4.0%)	530 (3.5%)	261 (3.0%)	139 (3.1%)
C	2,169 (10%)	1,720 (11%)	990 (12%)	548 (12%)
D	12,139 (58%)	8,609 (57%)	4,815 (56%)	2,508 (55%)
6-minwalk distance (feet)				
0	1,200 (5.8%)	813 (5.4%)	434 (5.1%)	243 (5.4%)
1-500	4,927 (24%)	3,455 (23%)	1,940 (23%)	995 (22%)
501-1,000	8,428 (41%)	6,153 (41%)	3,472 (41%)	1,805 (40%)
1,001-1,500	5,414 (26%)	4,098 (27%)	2,353 (28%)	1,264 (28%)
1,500+	656 (3.2%)	512 (3.4%)	312 (3.7%)	178 (4.0%)
Not assessed	138	115	77	53
Respiratory support				
Mechanical ventilation	1,565 (7.5%)	1,035 (6.8%)	536 (6.2%)	280 (6.2%)
BiPAP	1,914 (9.2%)	1,430 (9.4%)	786 (9.2%)	392 (8.6%)
CPAP	1,084 (5.2%)	726 (4.8%)	365 (4.3%)	157 (3.5%)
None	16,200 (78%)	11,955 (79%)	6,901 (80%)	3,709 (82%)
Cardiac index	2.80 (2.40, 3.28)	2.80 (2.40, 3.28)	2.81 (2.42, 3.30)	2.83 (2.45, 3.30)
Not assessed	1,956	1,521	960	557
Supplemental oxygen (L/min)	5 (3, 8)	4 (3, 8)	4 (3, 8)	4 (3, 8)
Performs ADLs[a] without assistance	1,081 (5.2%)	848 (5.6%)	503 (5.9%)	279 (6.1%)
Performs ADLs[a] with total assistance	1,764 (8.5%)	1,211 (8.0%)	598 (7.0%)	307 (6.8%)
Creatinine	0.80 (0.69, 0.98)	0.80 (0.68, 0.96)	0.80 (0.67, 0.95)	0.80 (0.66, 0.93)
Not assessed	1	1	1	
Sarcoid (group A)	220 (1.1%)	152 (1.0%)	98 (1.1%)	43 (0.9%)
Sarcoid (group D)	346 (1.7%)	253 (1.7%)	167 (1.9%)	93 (2.0%)
Lymphangioleiomyomatosis	79 (0.4%)	66 (0.4%)	45 (0.5%)	26 (0.6%)
Bronchiolitis obliterans	155 (0.7%)	121 (0.8%)	68 (0.8%)	30 (0.7%)
Female sex	8,305 (40%)	6,112 (40%)	3,482 (41%)	1,850 (41%)
Race/ethnicity				
Hispanic	1,533 (7.4%)	1,067 (7.0%)	554 (6.5%)	273 (6.0%)
Non-Hispanic Black	1,884 (9.1%)	1,344 (8.9%)	750 (8.7%)	384 (8.5%)
Non-Hispanic White	16,785 (81%)	12,350 (82%)	7,069 (82%)	3,770 (83%)
Other	561 (2.7%)	385 (2.5%)	215 (2.5%)	111 (2.4%)
Area deprivation index quintile				
Q1 (Lowest deprivation)	6,663 (32%)	4,846 (32%)	2,721 (32%)	1,451 (32%)
Q2	4,070 (20%)	3,000 (20%)	1,717 (20%)	901 (20%)
Q3	3,758 (18%)	2,777 (18%)	1,591 (19%)	824 (18%)
Q4	3,349 (16%)	2,446 (16%)	1,389 (16%)	762 (17%)
Q5 (Highest deprivation)	2,775 (13%)	1,987 (13%)	1,123 (13%)	575 (13%)
Not assessed	148	90	47	25

(Continued)

TABLE 3] *(Continued)*

Characteristic	Time of Transplant	1 Y	3 Y	5 Y
Insurance				
Private	10,383 (50%)	7,774 (51%)	4,667 (54%)	2,573 (57%)
Medicare	8,118 (39%)	5,768 (38%)	3,064 (36%)	1,516 (33%)
Medicaid	1,503 (7.2%)	1,078 (7.1%)	574 (6.7%)	293 (6.5%)
Other	759 (3.7%)	526 (3.5%)	283 (3.3%)	156 (3.4%)
Education				
High school or less	8,017 (40%)	5,910 (40%)	3,288 (37%)	1,688 (39%)
College/technical school	9,915 (50%)	7,236 (49%)	4,144 (50%)	2,198 (51%)
Postgraduate	2,080 (10%)	1,476 (10%)	823 (10%)	438 (10%)
Not assessed	751	524	333	214

Time points assessed include at the time of transplant, 1 year, 3 years, and 5 years. ADLs = activities of daily living; Group A = obstructive lung disease; Group B = pulmonary vascular disease; Group C = cystic fibrosis and immunodeficiency disorders; and Group D = restrictive lung diseases.
[a]No. (%); Median (IQR).

compared with WL survival. Subgroup-level model estimates were generally similar except when considering insurance status, where there was greater variability among recipients insured with Medicaid.

Discussion

The ability of prediction models to accurately discriminate lung transplant candidates most at risk of WL mortality from others on the list is essential to ensuring equitable access to transplant. Policies have prioritized the identification of those whose clinical condition is most urgent and whose predicted survival is greatest after transplant. Studies of the LAS system have suggested that using alternative modeling strategies, such as machine learning algorithms, and more contemporary data will improve the performance of WL and PT survival models[10]; we show these strategies do not achieve these important goals.

Using a contemporary retrospective cohort of lung transplant candidates, we evaluated the performance of the recently updated LAS WL model that will be used in the upcoming CAS system and compared them with six models designed to address potential challenges to model performance in this population. The presence of potential nonlinear relationships, variable interactions, and time-varying relationships could lead to predictive inaccuracy in the current LAS model. These elements were chosen because of the known presence of nonlinearity in some LAS variables, interrelatedness of variables that predict WL mortality, and the potential for changing relationships over time. Furthermore, we employed multiple machine learning techniques specifically designed to detect arbitrarily complex interactions among predictor variables—random survival forests.[11] We applied machine learning algorithms using existing modeling paradigms in which static predictions are used based on the most recent available data; future applications of machine learning may be well suited to using an entire range of illness trajectory to improve predictive abilities of models.

The AUC was used to assess model discrimination or the ability of the LAS to predict WL removal or death in candidates on the lung transplant waiting list. Thus, discrimination indicates how well the LAS performs in its risk estimates compared with the number of events that occur.[12] All WL models had high discrimination performance, which decreased as individuals spent more time on the waiting list. Differences existed between models, but no single model clearly outperformed the others. In particular, accounting for nonlinearity, interactions, or time-varying relationships did not lead to significant improvements in predictive performance of WL mortality model estimates.

The PT models predicted only slightly better than chance (AUC range, 0.59-0.61) and additional modeling efforts failed to yield improvements. This phenomenon has been previously described, and the PT survival model is inherently limited by the

TABLE 4] Discrimination Performance of PT Survival Models

Baseline Cohort							
Forecasted Time	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII
1 Y	0.61 (0.57-0.65)	0.58 (0.54-0.63)	0.60 (0.56-0.64)	0.59 (0.55-0.63)	0.59 (0.57-0.62)	0.59 (0.57-0.62)	0.60 (0.57-0.62)
3 Y	0.60 (0.57-0.64)	0.59 (0.55-0.63)	0.60 (0.56-0.63)	0.59 (0.55-0.62)	0.58 (0.56-0.61)	0.58 (0.56-0.61)	0.59 (0.57-0.62)
5 Y	0.60 (0.57-0.63)	0.60 (0.57-0.62)	0.60 (0.57-0.63)	0.60 (0.57-0.63)	0.59 (0.57-0.62)	0.59 (0.57-0.62)	0.59 (0.56-0.61)
1 Y Cohort							
Forecasted Time	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII
Forecasted Time Ahead							
1 Y	0.59 (0.55-0.62)	0.57 (0.53-0.61)	0.58 (0.54-0.62)	0.57 (0.54-0.61)	0.56 (0.53-0.59)	0.56 (0.53-0.59)	0.57 (0.55-0.60)
3 Y	0.58 (0.55-0.61)	0.58 (0.56-0.61)	0.59 (0.56-0.62)	0.57 (0.54-0.60)	0.57 (0.55-0.59)	0.57 (0.55-0.59)	0.58 (0.55-0.60)
3 Y cohort							
Forecasted Time	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII
Forecasted time ahead							
1 Y	0.56 (0.52-0.61)	0.57 (0.53-0.62)	0.57 (0.52-0.63)	0.56 (0.49-0.63)	0.56 (0.51-0.59)	0.56 (0.51-0.59)	0.56 (0.52-0.60)
3 Y	0.57 (0.54-0.61)	0.60 (0.55-0.64)	0.59 (0.55-0.64)	0.59 (0.54-0.65)	0.58 (0.55-0.62)	0.58 (0.54-0.62)	0.59 (0.55-0.63)
5 Y cohort							
Forecasted Time	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII
Forecasted time ahead							
1 Y	0.57 (0.51-0.63)	0.59 (0.52-0.66)	0.60 (0.52-0.67)	0.59 (0.50-0.68)	0.60 (0.54-0.65)	0.60 (0.54-0.65)	0.59 (0.53-0.68)
3 Y	0.55 (0.50-0.61)	0.62 (0.56-0.68)	0.62 (0.55-0.69)	0.62 (0.56-0.68)	0.62 (0.56-0.67)	0.62 (0.56-0.67)	0.61 (0.56-0.68)

Data are median area under the curve (min, max). Model I: PT-CAS; Model II: Re-estimation of PT-CAS to the specified residual cohort with an updated cohort and parameterizations; Model III: A modification of Model II with cubic splines accounting for nonlinear relationships; Model IV: Random survival forest model; Model V: A logistic model calculated using the stacked survival method; Model VI: Linear discriminant analysis; Model VII: Gradient boosted tree model. The forecasted time ahead indicates the time from the cohort-defined period. For example, forecasted time ahead of 1 year from the baseline cohort would be 1 year in actual time, whereas 1 year from the 3-year cohort would be 4 years in actual time. CAS = composite allocation score; PT = posttransplant.

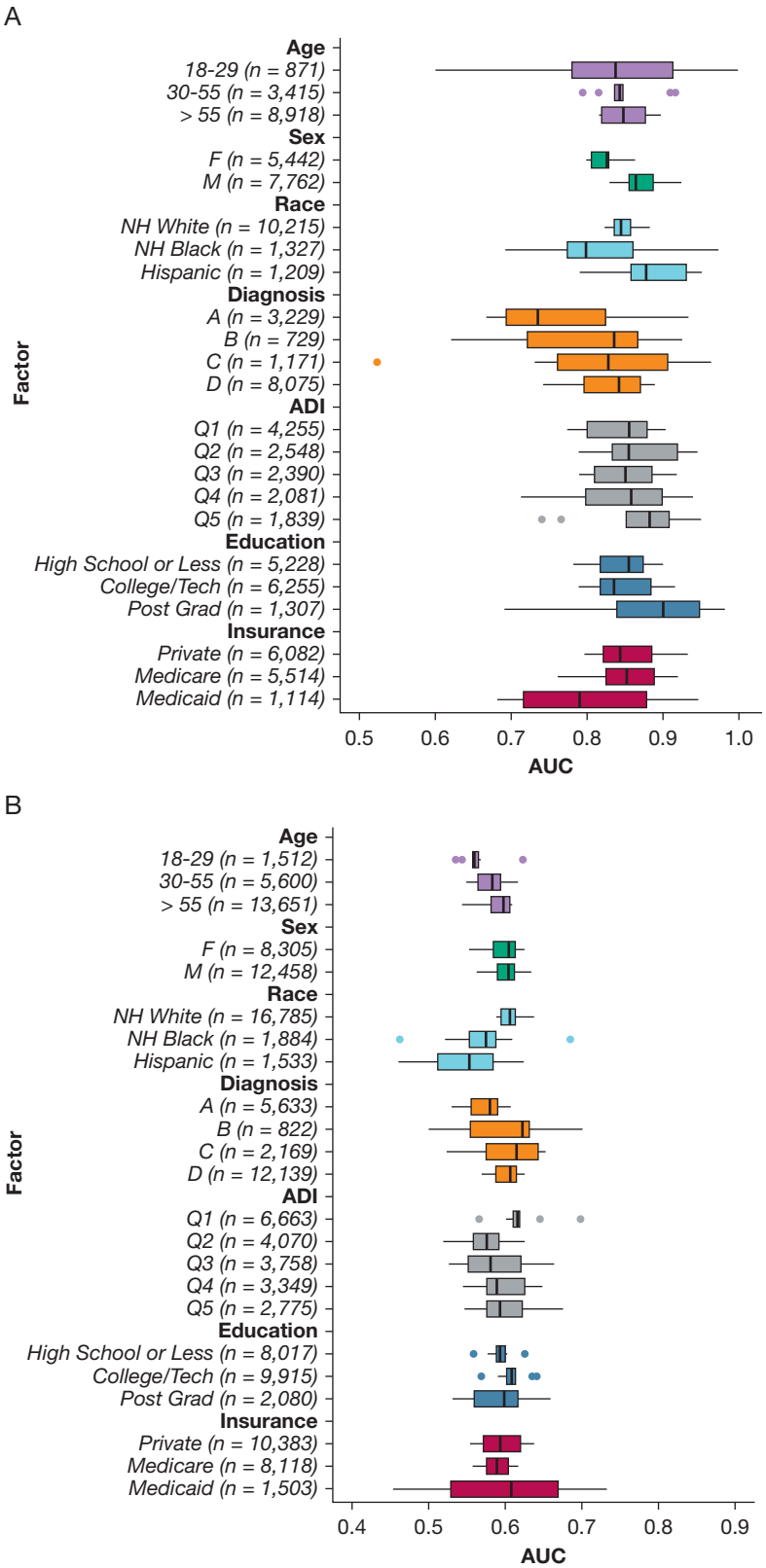

Figure 1 – *Comparison of discrimination performance across socioeconomic and clinical factors. Boxplots of 10-fold cross-validation for AUC within sociological and demographic factors for the WL-CAS model at 1 year (A) and PT-CAS at 5 years from transplant (B) (n =) indicates the total sample size across 10 folds. ADI = area deprivation index; AUC = area under the curve; Group A = obstructive lung disease; Group B = pulmonary vascular disease; Group C = cystic fibrosis and immunodeficiency disorders; and Group D = restrictive lung diseases; NH = non-Hispanic; PT = posttransplant; WL = waitlist.*

inclusion of only those variables that are known at the time of allocation.[13] Additionally, the population of transplant recipients is more homogeneous than the WL population, which makes accurately characterizing outcomes more difficult. Finally, factors known to limit PT survival, including perioperative

factors, early postoperative factors, and transplant-specific events such as the development of acute and chronic rejection, cannot be known at the time of allocation.[14-16] Discrimination degraded only slightly when accounting for cohorts based on conditional survival.

Subgroup analysis suggested lower discrimination performance of the WL mortality model for females (vs males), non-Hispanic Black candidates (vs candidates of other races or ethnicities), group A candidates (vs other diagnosis groups), Medicaid candidates (vs those with other types of insurance), and those with lower levels of education (vs those with postgraduate education). Although discrimination performance of the PT models was generally poor, comparisons across subgroups suggested that PT survival model discrimination was lower for non-Hispanic Black and Hispanic recipients (vs non-Hispanic White recipients), younger (vs older) recipients, those from areas with higher levels of socioeconomic deprivation (ADI quintiles 2-5 vs ADI quintile 1), and group A recipients, but there were no clear relationships with respect to education and insurance type. In other words, our analysis suggests that current and future risk-based lung allocation policies are dependent on assessments of acuity and benefit that are systematically less accurate for vulnerable populations. Our analysis cannot explain why these differences occurred, and therefore the reasons for these disparities should be further explicated.

Interpretation

We found that discrimination performance for the current LAS and forthcoming CAS WL and PT survival models could not be improved using contemporary data or more advanced statistical modeling techniques. It is therefore our assessment that existing allocation models have maximized predictive capabilities of the available risk factors currently captured by the OPTN. Further investigation into the accuracy of current models in prospective cohorts as well as the prognostic contribution of risk factors that are external to existing models, including recipient comorbidities and donor characteristics at the time of allocation, are needed.

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Additional information: The e-Appendix, e-Tables and e-Figure are are available online under "Supplementary Data".

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